Thickness from Color by Inverting the Physics: Self-Calibrating Differentiable Thin-Film Optics for Region-Level 2D-Material Thickness Estimation

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Abstract

Estimating the thickness of exfoliated two-dimensional flakes from ordinary optical micrographs is a foundational task in 2D-material fabrication. Current learning systems use thin-film optics only offline, as a synthetic-data generator for networks that classify thickness into discrete bins, and devote substantial machinery to closing the resulting gap between simulated and real imaging conditions. We instead place a differentiable transfer-matrix model of the full optics-to-color pipeline *inside* the inference loop, recovering thickness as a continuous variable by profile-likelihood continuation jointly with the imaging nuisances (oxide thickness, sensor gains, illuminant spectrum). The bare substrate present in every micrograph then acts as an in-frame calibration target, replacing domain adaptation with self-calibration and requiring no labeled training images. On the public micrographs evaluated here that mechanism is only partially available — three free channel gains reproduce any substrate colour exactly, so the substrate term is uninformative and we fall back to a restricted mode with a pinned illuminant and one oxide calibrated per dataset — and diagnosing that is itself one of the real-data results. The method requires no labeled training images (the illuminant basis is built from published measured spectra, which we treat as calibration data). On held-out *simulated* scenes with adversarially misspecified illuminants, median absolute error is 0.104 nm for graphene and 0.154 nm for MoS_2 with zero retraining, since supporting a new material is one dispersion table; on simulated flakes of physically realistic integer-layer thickness it counts layers with 82.2% exact and 100% within-one-layer accuracy and resolves monolayers to 0.015 nm, an inferred-thickness error of about 4% of a graphene layer. These are simulation figures; the corresponding real-data evidence is the recovered interlayer spacing reported below, not a per-flake accuracy. Cramér–Rao analysis establishes what is extractable from three color channels, shows that the problem is unidentifiable without priors, and predicts the residual failure domain, which we characterize as joint thickness–nuisance ridges rather than classical color metamers and address with two inexpensive acquisition protocols, evaluated on simulated scenes. We also show that the objective's numerical aperture must be modeled: a normal-incidence approximation is harmless at NA 0.55 but destroys thin-regime accuracy at NA 0.9. On real micrographs from an external laboratory (the MaskTerial graphene substrate-variance series, n=200 held-out flakes each) a restricted mode using no task-specific labeled training data attains 95.3% exact layer agreement at n=600 and recovers graphene's interlayer spacing in statistical agreement with its independently known value, along with each dataset's effective substrate thickness, from the images alone. Code and experiments are released.

1 Introduction

Research on two-dimensional materials begins, in practice, at an optical microscope. Mechanically exfoliated flakes of graphene, hexagonal boron nitride, and transition-metal dichalcogenides are scattered across oxidized silicon wafers, and a human or a computer-vision system must find the rare thin flakes and estimate how many atomic layers they contain. The physical signal is thin-film interference: light reflected from the air/flake, flake/oxide, and oxide/silicon interfaces interferes with a thickness-dependent phase, so thickness modulates apparent color [1, 2]. This is why the community standardizes on SiO_2/Si wafers with particular oxide thicknesses chosen to maximize few-layer visibility.

A recent line of work automates the task with deep learning. MaskTerial [3] couples an instance-segmentation network, pre-trained on physics-based synthetic images, to a mixture-model classifier over optical contrast that assigns flakes to discrete layer classes; adapting to a new material requires 5–10 labeled real images. φ-Adapt [4] frames the problem as physics-informed domain adaptation: a transfer-matrix simulator generates synthetic source-domain images, learned modules estimate normalized color and a single-channel reflectance proxy, and thickness is predicted by a quantized classifier with a regression head, reaching a mean error of 5.8 nm on 177 flakes of 10–240 nm that the authors collected themselves. CLIFF [5] attacks the same data-scarcity problem from a third direction, framing layer classification as continual learning that acquires new materials incrementally while retaining earlier ones through memory replay with knowledge distillation.

Most recently, QuPAINT [6] extends the same program to multimodal language models: a physics-based generator synthesises flake images, an instruction dataset supplies physics-informed supervision, and a Physics-Informed Attention module fuses visual embeddings with optical priors. That work is explicit about why it stops short of inverting the optics, stating that a full transfer-matrix simulation would require knowledge of wavelength-dependent refractive indices, illumination spectra and camera responses that is typically unavailable or varies between laboratories, and adopting a perceptual CIE LAB color-distance proxy instead. We take that statement as an accurate description of the obstacle and as the motivation for this paper: those quantities need not be known in advance, because the bare substrate in the same frame determines them.

All of these systems use the physics in the same restricted way: the transfer-matrix method (TMM) runs *forward, offline, once*, to manufacture training data, after which a network is asked to memorize the inverse map and a second apparatus (adaptation, fine-tuning, continual learning) is asked to patch the mismatch between simulated and real imaging conditions. Neither the inversion nor the calibration uses the physics at test time. Meanwhile the TMM has been made differentiable [7, 8] and is routinely used for gradient-based *design* of optical coatings; to our knowledge it has not been used as an inference-time decoder for flake metrology.

Scope. We address thickness estimation *given segmented regions*: our input is a flake region and a bare-substrate region, not a raw micrograph. Segmentation is a substantial part of an end-to-end system (it is the first stage of MaskTerial), and we do not attempt it. Comparisons to end-to-end systems in this paper therefore favor us, and we say so wherever such a comparison appears; §5.7 quantifies what imperfect segmentation costs.

This paper treats flake thickness estimation as analysis-by-synthesis through differentiable physics. A forward model maps thickness d, oxide thickness t_{ox}, per-channel sensor gains g, and illuminant spectral-shape coefficients c to the linear-RGB value of a pixel via the TMM, the illuminant spectrum, and the CIE observer. Inference is maximum-a-posteriori estimation by *profile-likelihood continuation*: thickness is swept over a dense grid while the nuisance subproblem is solved by warm-started Levenberg–Marquardt, tracing the exact profile posterior and enumerating every thickness mode by construction; candidates are then polished by joint optimization through the physics. Nuisances are estimated jointly with thickness, anchored by the bare substrate that occupies most of every real micrograph and whose reflectance is computable from first principles. This single design choice replaces the field's domain-adaptation problem with a self-calibration problem that the image itself solves.

Contributions. (i) To our knowledge the first use of differentiable thin-film physics as an inference-time decoder for 2D-flake thickness from color microscopy, recovering thickness as a continuous variable rather than a class label, with a continuation-based global inference scheme (§3). Model-based inversion itself is long established in optical metrology (§2); what is new here is the modality, the self-calibration, and the identifiability analysis it requires. (ii) An identifiability theory for RGB-to-thickness inversion: Cramér–Rao bounds with and without known imaging conditions, quantifying self-calibration, and an empirically supported account of the residual ambiguity as joint thickness–nuisance ridges rather than classical metamers (§4, §4.1). (iii) A held-out benchmark under nuisance corruption, with a metric- and range-matched comparison to the state of the art, an ablation against simpler methods including an oracle-calibrated upper bound, and a per-scene analysis showing that the estimator is not the limiting factor on accuracy (§5). (iv) Acquisition protocols, derived from the theory, that substantially reduce the remaining failure mode: multi-region joint inference and a known-filter two-frame protocol with a profile-triggered acquisition rule (§5.8). (v) Validated oblique-incidence/NA-averaged optics, showing that modeling the objective's numerical aperture is a requirement rather than a refinement (§5.9), plus robustness and negative results (§5.10, §5.11).

2 Related work

Optical thickness identification of 2D materials. Quantitative contrast methods date to the first years of graphene research [1, 2]: Fresnel/TMM modeling of the air/flake/SiO_2/Si stack predicts contrast versus thickness, and measured contrast under narrow-band illumination counts layers. Learning-based systems scaled this to automated wafer search [14, 15], culminating in the physics-informed synthetic-data systems discussed above [3, 4, 5, 6]. All perform discrete classification over layer bins, use physics offline only, and require adaptation data or machinery per deployment. Our method differs on each axis: continuous estimation, physics in the loop, and calibration from the test image itself.

Model-based optical metrology. Fitting a transfer-matrix forward model to measured optical signals, with nuisance parameters and prior regularization, is long-established practice in semiconductor metrology: spectroscopic ellipsometry [16, 17] and scatterometry [18] both invert physical stack models against measurements, and the statistical apparatus we use (Fisher information, Cramér–Rao and hybrid Bayesian bounds, profile likelihood, Levenberg–Marquardt) is standard [19, 20, 21, 22, 23]. Our contribution is not the idea of model-based inversion. It is (i) that the measurement is three broadband color channels from a commodity microscope camera rather than a spectrometer or ellipsometer, which makes identifiability the central question rather than a formality; (ii) that the bare substrate present in every micrograph supplies in-frame calibration, removing the per-laboratory calibration step those methods assume; (iii) the resulting identifiability analysis and ridge geometry (§4–4.1); and (iv) acquisition protocols derived from that analysis (§5.8). Relative to the 2D-materials literature, the novelty is using the physics at inference time at all.

Differentiable optics and spectral fitting. Differentiable TMM implementations [7, 8] enable gradient-based inverse *design* of multilayer coatings. Spectroscopic ellipsometry and reflectometry solve per-point thickness inversion by fitting measured *spectra* to the TMM, but require spectrally resolved hardware. Our setting is harsher and more practical: three broadband color channels from a commodity camera, unknown illumination, and a per-region inverse problem. The identifiability analysis of §4 is, to our knowledge, the first formal treatment of when this 3-channel inversion is well-posed, and the two-frame filter protocol of §5.8 recovers coarse spectral leverage with commodity hardware.

Analysis-by-synthesis. Inverting a graphics-style forward model by optimizing latent scene parameters against an observed image is a classical idea enjoying a renaissance under differentiable rendering. We instantiate it with a wave-optics forward model whose recovered parameters are physically meaningful and independently falsifiable: an oxide thickness inferred by our method is a testable claim about the wafer.

3 Method

3.1 Forward model: from thickness to pixel value

Thin-film stack. At normal incidence, the air/flake(d)/SiO_2(t_{ox})/Si stack has reflectance computed by the characteristic-matrix TMM: each layer j with complex index $n_j(\lambda)$ and thickness d_j contributes

$$M_j = \begin{pmatrix} \cos\delta_j & \frac{i}{n_j}\sin\delta_j \\ i\,n_j\sin\delta_j & \cos\delta_j \end{pmatrix}, \qquad \delta_j = \frac{2\pi n_j d_j}{\lambda}, \tag{1}$$

and with $(B, C)^\top = \left(\prod_j M_j\right)(1, n_s)^\top$, $R(\lambda) = \left|\frac{n_0 B - C}{n_0 B + C}\right|^2$. Our JAX implementation agrees with the reference `tmm` package [7] to four decimal places at normal incidence and to machine precision at oblique incidence (§5.9); the downstream color pipeline (CMF integration, illuminant normalization and XYZ→linear-sRGB conversion) independently reproduces the `color-science` implementation to a relative 4×10^{-4}, below the noise floor, after locating and fixing a time-convention sign error whose only symptom was a graphene layer that *brightened* rather than darkened the substrate — the erroneous model's bare-substrate predictions agreed with the reference to three decimals, illustrating how silently such bugs pass coarse validation.

Materials. Si uses the Aspnes–Studna tabulated dielectric function [9]; SiO_2 the Malitson Sellmeier dispersion [10]; graphene the standard constant-index model $n = 2.6 - 1.3i$ [1], with layer height 0.335 nm used only to convert continuous thickness to layer counts when reporting, never inside the forward model or the inference; MoS_2 the tabulated ellipsometry of Song et al. [11]; hBN, used in §5.5, the ordinary-ray data of Grudinin et al. [12]. All tabulated data are drawn from the refractiveindex.info database [13]. Supporting an additional material requires its wavelength-dependent optical constants; no retraining, no labeled examples and no change to the inference procedure.

Color formation. A region of thickness d has linear-RGB value

$$I_{\mathrm{RGB}}(d, t_{\mathrm{ox}}, g, c) \;=\; g \odot \left[T \, \frac{\int L(\lambda; c)\, R(\lambda; d, t_{\mathrm{ox}})\, \bar{\mathbf{x}}(\lambda)\, d\lambda}{\int L(\lambda; c)\, \bar{y}(\lambda)\, d\lambda} \right], \tag{2}$$

with $\bar{\mathbf{x}}$ the CIE 1931 observer, T the XYZ→linear-sRGB matrix, $g \in \mathbb{R}^3_+$ per-channel gains absorbing white balance and exposure, and $L(\lambda; c) = L_{\mathrm{D65}}(\lambda) + \sum_{k=1}^{5} c_k B_k(\lambda)$. The basis $\{B_k\}$ is a PCA over eleven measured illuminants spanning incandescent/halogen, daylight D50–D75, LED, and fluorescent sources, luminance-normalized so overall intensity is carried by g; five components capture 99.0% of the family's aggregate variance, pooled over all eleven spectra; the mean *per-illuminant* reconstruction residual, as an RMS fraction of that spectrum's own magnitude, is 9.2%, the two differing because the pooled measure is dominated by the spectra of largest variance. A *physical positivity constraint* penalizes negative spectral power, $\max(-L, 0)$: diagnosis showed wrong interference orders exploiting unphysical negative-illuminant fits (minimum L down to -0.048), and this constraint, which requires no additional measurement, removes that escape route. The penalty is non-smooth at $L = 0$, which Levenberg–Marquardt does not formally accommodate; it is one-sided and inactive wherever the fitted spectrum is non-negative, so it enters the local quadratic model only on iterates that have already left the physical region, and the solver diagnostics of §3.2 were run with it active. The entire map (2) is differentiable in $(d, t_{\mathrm{ox}}, g, c)$.

3.2 Inference: profile-likelihood continuation with joint polishing

Let $\theta = (\log d,\, t_{\mathrm{ox}},\, \log g,\, c) \in \mathbb{R}^{10}$. Observations are region means $\bar{y}_f$ (N_f flake pixels) and $\bar{y}_b$ (N_b bare-substrate pixels) — sufficient statistics under i.i.d. Gaussian pixel noise σ — and the negative log posterior is

$$\mathcal{L}(\theta) = \frac{N_f}{2\sigma^2}\big\|\bar{y}_f - I_{\mathrm{RGB}}(d,\cdot)\big\|^2 + \frac{N_b}{2\sigma^2}\big\|\bar{y}_b - I_{\mathrm{RGB}}(0,\cdot)\big\|^2 + \frac{(t_{\mathrm{ox}} - t_0)^2}{2\sigma_t^2} + \frac{\|\log g\|^2}{2\sigma_g^2} + \sum_k \frac{c_k^2}{2\sigma_{c,k}^2} + \rho \sum_\lambda \big[-L(\lambda; c)\big]_+^2, \tag{3}$$

with physically grounded priors $t_0{=}285\,\mathrm{nm}$, $\sigma_t{=}5\,\mathrm{nm}$ (wafer spec), $\sigma_g{=}0.1$, $\sigma_{c,k}$ the empirical illuminant-family spread, and the final term the spectral positivity penalty. The bare-substrate term is the self-calibration mechanism: it ties (t_{ox}, g, c) to a spectrum computable from first principles. Because the problem is unidentifiable without priors (§5), these priors are load-bearing, and this imposes an operational requirement: the nominal oxide thickness must be known to roughly $\pm 15\,\mathrm{nm}$, which any wafer specification provides.

In our experiments the dominant multimodality lies along d, on interference-order branches, while the nuisance subproblem at fixed d was empirically a benign nonlinear least squares; the nuisance geometry is not globally benign, as the continuation hysteresis and wrong-branch tracking described below show. We therefore (1) sweep d over a dense log grid, solving the nuisance subproblem at each point with Levenberg–Marquardt warm-started from the neighboring point — *bidirectionally*, and with a cold prior-mean companion solve at every point, both verified necessary: one-way warm starts exhibit continuation hysteresis (profile discrepancies up to 10^4 posterior units), and warm-only sweeps were observed tracking wrong *oxide* interference branches with 12σ prior violations; (2) extract candidate modes as local minima of the profiled posterior within a wide window (40 units) of the best; (3) polish every candidate with full joint LM through the physics and answer with the lowest *polished* posterior. The two-tier design (wide window for candidacy, polished loss for selection) matters: the true mode can sit 12–40 units above the coarsely-solved profile minimum and win only after polishing; adopting it cut hard-regime failures from 5/25 to 1/25 with, verified pairwise, zero changes on 100 standard-benchmark scenes. Inference takes $\sim$1.5 s per scene on CPU.

Multi-region extension (mode-lattice inference). Real exfoliated samples contain multiple thickness regions imaged identically. We run the single-region profile per region (exhaustive per-region mode enumeration), form the lattice of joint hypotheses ($\leq 4 \times 4$ modes), and polish every lattice point under *one shared nuisance set*. The construction assumes the joint optimum is a product of the two regions' marginal modes; we checked this against a dense two-dimensional search over (d_1, d_2) with nuisances re-optimized at every grid point, and the lattice recovered the grid optimum to within 10^{-2} posterior units on every scene tested. The mechanism is that a nuisance vector that rescues the wrong interference order for one region must survive the other region's data, and every order combination is explicitly tested rather than hoped-for by random restarts.

4 Identifiability theory

Oracle bound. With nuisances known, the single-pixel Cramér–Rao bound at $\sigma = 0.5\%$ pixel noise is 0.093 nm at monolayer graphene on 285 nm oxide, with median 0.68 nm over $d \in [0.335, 120]$ nm and $t_{\mathrm{ox}} \in [50, 350]$ nm; **99.2% of that grid lies below the 5.8 nm error of the current state of the art** [4]. We state the comparison carefully, because the two quantities are not of the same kind: 0.68 nm is a *single-pixel* bound with nuisances known, computed under our measurement model, while 5.8 nm is a published empirical mean error over regions. The ratio of 8.5× therefore illustrates headroom relative to a modeled information floor rather than an efficiency deficit of that method. Averaging raises the floor's stringency considerably: at the 100 flake and 300 background pixels used throughout the rest of this paper the known-nuisance bound falls to 0.029 nm (§4, below). As a byproduct, the contrast-optimal oxide thickness is strongly definition-dependent, varying with both wavelength (88/276 nm at 550 nm, 84/265 at 530, 96/301 at 600, and 79/224 nm for broadband RGB-norm contrast) and, more importantly, with numerical aperture (§5.9): at the NA 0.55–0.9 objectives actually used for flake hunting the optima move to 90/286–96/304 nm, recovering the community's standard wafers. Blake et al. [1] make the wavelength dependence explicit, noting that graphene's visibility depends strongly on both oxide thickness and illumination wavelength. Quantitative methods should state which contrast definition and which aperture they assume.

Honest bound: nuisances estimated, not known. With Fisher information $F = \sigma^{-2}(N_f J_f^\top J_f + N_b J_b^\top J_b) + F_{\mathrm{prior}}$, the marginalized bound on d (Fig. 2) shows: (a) *self-calibration is affordable, and we can now price each nuisance separately.* Computed through a single code path at 100 flake and 300 background pixels, the median bound over $d \in [0.335, 120]$ nm degrades from 0.029 nm with all nuisances known, to 1.29× that when oxide and gains are unknown, 5.96× when a three-component illuminant is added, and 7.80× for five components. At monolayer the corresponding factors are 1.19, 1.96 and 2.02. The ordering is the substantive result: geometric and radiometric nuisances are nearly free, whereas the illuminant *spectrum* costs a factor of five and is the dominant term. Bounds converge toward the oracle floor as background evidence accumulates. The illuminant is thus the dominant nuisance, which motivates the two-frame protocol of §5.8; what φ-Adapt approximates with learned normalization modules and test-time entropy minimization is here an explicit, quantified cost; (b) the penalty is *non-monotonic in thickness* (1.73× at 5 nm, 1.02× at 20 nm, 2.38× at 100 nm), predicting where domain shift should hurt classification baselines; (c) the thickness-to-color map is non-injective (Fig. 1, right), bounding *any* single-frame method in the thick regime.

4.1 The residual ambiguity is a joint ridge, not a metamer

The failures that survive all single-frame improvements were assumed, by us and implicitly by the oracle analysis, to be classical metamers: two thicknesses producing the same color. Direct examination falsified this. On confidently-wrong scenes, the true thickness is *not* close in color to the estimate at the fitted nuisances (gain-invariant ratio distances 0.08–0.27 against a noise scale of ∼0.006, with no rival minimum near the truth). Instead, the nuisance parameters drift *continuously and jointly with thickness* along a ridge until the wrong interference order fits the frame essentially perfectly. Three natural mitigations fail for exactly this reason, and we verified each: ambiguity detection by counting posterior modes (the wrong solutions look unimodal — 0 of 5 failures flagged); rival search by fixed-nuisance metamer scanning (finds nothing near the truth); and discrimination scores computed from fitted-mode color predictions (§5.11). The operative object is the *profile posterior* $L^*(d)$, in which nuisances are re-optimized per thickness: the true solution appears there as a rival basin (typically 2–25 units above the wrong winner), which is what makes profile-based candidacy, triggering, and reporting sound. We advance this as a hypothesis with converging

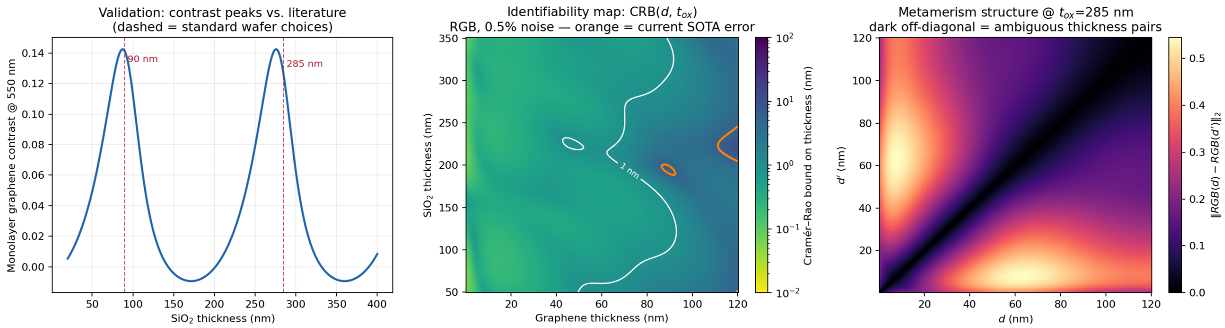

Figure 1: Forward-model validation and oracle identifiability. *Convention:* this figure reports single-pixel oracle bounds (nuisances known), used only to establish the information floor of the measurement; all bounds quoted in the text and in Fig. 2 are prior-augmented and marginalized over the full nuisance set at 100 flake and 300 background pixels. **Left:** monolayer graphene contrast vs. oxide thickness (peak 0.142); dashed lines mark standard wafers. **Center:** oracle CRB over (thickness, oxide); orange contour = 5.8 nm SOTA error, white = 1 nm. **Right:** pairwise color distance at t_{ox}=285 nm; dark off-diagonal bands are ambiguous thickness pairs in the oracle setting.

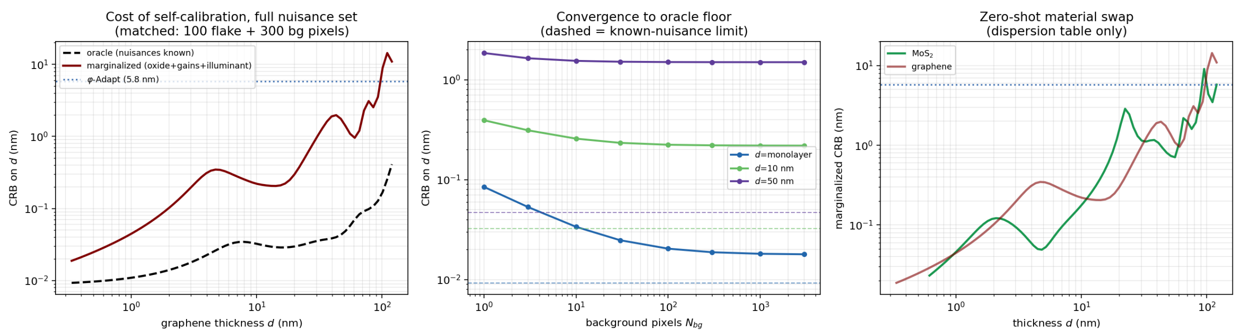

Figure 2: The honest bound, computed with authoritative optical constants and the full nuisance set (oxide, gains, 5-component illuminant), matched at 100 flake and 300 background pixels. **Left:** oracle vs. fully marginalized CRB for graphene; the gap is the quantified price of self-calibration. **Center:** convergence toward the known-nuisance floor (dashed) as background evidence accumulates. **Right:** MoS_2 after a dispersion-table swap.

support rather than a settled theory: it rests on direct inspection of five failures plus the falsification of three mitigations it predicts should fail, and a larger diagnostic study would strengthen it.

5 Experiments

5.1 Benchmark design

Scenes are constructed to be *hostile*: $d \sim$ log-uniform over $[1\ \text{layer}, 120\,\text{nm}]$; $t_{\mathrm{ox}} \sim \mathcal{N}(285, 5^2)$; $\log g \sim \mathcal{N}(0, 0.1^2)$; pixel noise 0.5%; N_f=100, N_b=300; and the illuminant drawn from the *full eleven-spectrum family* (including spiky fluorescents) while the estimator has only the 5-component basis — 1% of illuminant variance is unmodelable by construction, so the benchmark contains genuine model misspecification rather than an inverse crime. We state plainly what this benchmark does *not* test: oxide, gains, and noise are drawn from the estimator's own priors, so it is a well-specified Bayesian setting in those coordinates. We therefore add a **prior-violating stress test** in which oxide dispersion is 3× the assumed prior (σ=15 nm), gain dispersion 2.5× ($\sigma_{\log g}$=0.25), and noise 2× the assumed level: held-out median 0.203 nm, mean 1.31 nm, failures 5/60 (8.3%), sub-25 nm median 0.100 nm (n=60, seed 205). Degradation is mild and self-calibration continues to recover the true oxide despite the mislabeled prior.

5.2 Main results

Protocol. To avoid selection effects, all design choices (mode window, positivity weight, basis size, solver schedule, grid density) were frozen after development on seeds $\{0, 3, 7, 11-13, 21, 22, 31-34, 55, 77, 99\}$, and the numbers below were then produced on *held-out* seeds $\{201-207\}$ never used for any decision. Held-out and development performance agree closely (median 0.104 vs. 0.123 nm), indicating no overfitting to the development scenes.

Table 1 and Fig. 3 report results. Our method's median appears at three values across this section — 0.104 nm (Table 1, seeds 201–202, n=120), 0.099 nm (Table 2, the seed-201 subset, n=60, used so that baselines are compared on identical scenes) and 0.130 nm (§5.7, seed 211, n=40) — because these are the same estimator on different held-out subsets; the spread indicates the sampling variability to expect at these sample sizes. Median absolute error is **0.104 nm** for graphene (n=120) and **0.154 nm** for MoS_2 (n=80) with no retraining, fine-tuning, or labeled data of any kind — the material swap is one dispersion table, against MaskTerial's 5–10 labeled images per material [3]. In the sub-25 nm regime, 1 of 148 held-out estimates failed (0/93 graphene; 1/55 MoS_2, where d=10.4 nm was reported as 74.8 nm).

Comparison to the state of the art, stated carefully. φ-Adapt reports a 5.8 nm *mean* error over 10–240 nm [4], measured on 177 real flakes the authors collected for that purpose. That set is not public, so their number cannot be reproduced or re-run by us on identical data; this is a limit on the comparison that no amount of care on our side removes. Restricting our held-out scenes to the overlapping range ($d \geq 10$ nm, n=47) and using their metric, our mean error is 2.86 nm: a **2.0**× improvement. Other metric and range choices give much larger multipliers (full-range mean 4.9×; full-range median 56×), but we regard the range- and metric-matched figure as the only defensible headline. Even this restriction favors us: their range extends to 240 nm while ours stops at 120 nm, and the excluded band is precisely where our single-frame method is ridge-limited. Three caveats apply and none is minor: their number is measured on real images and ours on synthetic scenes; their pipeline includes segmentation error while ours assumes given regions; and their range extends beyond ours into the thick regime where our single-frame method is ridge-limited. §5.12 is the experiment that removes the first two.

Identifiability and efficiency. Two observations, made against per-scene bounds evaluated at each scene's *true* parameters. First, the *frequentist* CRB is unbounded. Six numbers (two regions × three channels) cannot determine ten parameters, so the Fisher matrix is rank-deficient; that alone would not make the *scalar* bound on d infinite, since a rank-deficient matrix can still leave one coordinate estimable. It is infinite here because the null directions are not orthogonal to d: the ridge geometry of §4.1 is precisely a family of nuisance excursions that move thickness while leaving the six observations unchanged. The priors of (3) are therefore not a convenience but the reason the problem is well posed at all. Second, against a hybrid bound formed by adding the negative Hessian of the log prior at the true parameter to the likelihood Fisher information — a local regularized diagnostic rather than an exact Van Trees bound, since the positivity constraint is not of that form — achieved errors are at or below the bound — median |error|/CRB of 0.28 under real measured illuminants (n=35, thin regime) and 0.88 when the illuminant is drawn from inside the estimator's basis span (n=35) — versus ≈ 0.67, the value an efficient unbiased estimator would give if its errors were Gaussian — a reference point under that assumption, not a standard efficiency ratio. We therefore claim only that *the estimator is not the limiting factor*: a MAP estimator with informative priors is deliberately biased and the positivity constraint contributes information the linearized Fisher analysis omits, so sub-0.67 ratios are expected and this comparison is a diagnostic rather than a proof of efficiency. We flag that an earlier version of this analysis, which linearized the bound at illuminant coefficients recovered under a mismatched normalization (up to 327σ from the prior, i.e. at physically invalid spectra), produced ratios near 10 and supported the opposite conclusion; the corrected analysis supersedes it.

Robustness to illuminants outside the model family. The spectral basis is a PCA over eleven measured illuminants, and both it and the benchmark draw on that family, so we tested illuminants of entirely different physical type: blackbody sources at 2800/3400 K, a xenon arc, a narrow-band high-CRI LED, and a mercury arc with strong emission lines (26–39% of their spectral deviation is unrepresentable in the basis, against 9% for in-family sources). Thin-regime performance is essentially unaffected: median

Table 1: Held-out benchmark (frozen configuration, seeds never used in development). Failures are errors >5.8 nm; 95% CIs are Clopper–Pearson. "v1" is the multi-start first-order ablation of our own estimator (§5.11), evaluated on development seeds.

	Graphene (n=120)	MoS_2 zero-shot (n=80)	Graphene, v1 ablation
Median |error|	**0.104** nm	**0.154** nm	0.276 nm
Mean |error|	1.19 nm	1.95 nm	2.85 nm
Mean, $d \geq 10$ nm (SOTA range)	2.86 nm	—	—
Median, $d < 25$ nm	**0.055** nm	**0.073** nm	0.153 nm
Failures	8/120 (6.7%, CI [2.9, 12.7])	4/80 (5.0%, CI [1.4, 12.3])	17/120
Failures, $d < 25$ nm	0/93	1/55	0/90
Time / flake (CPU)	1.5 s	1.5 s	3.5 s

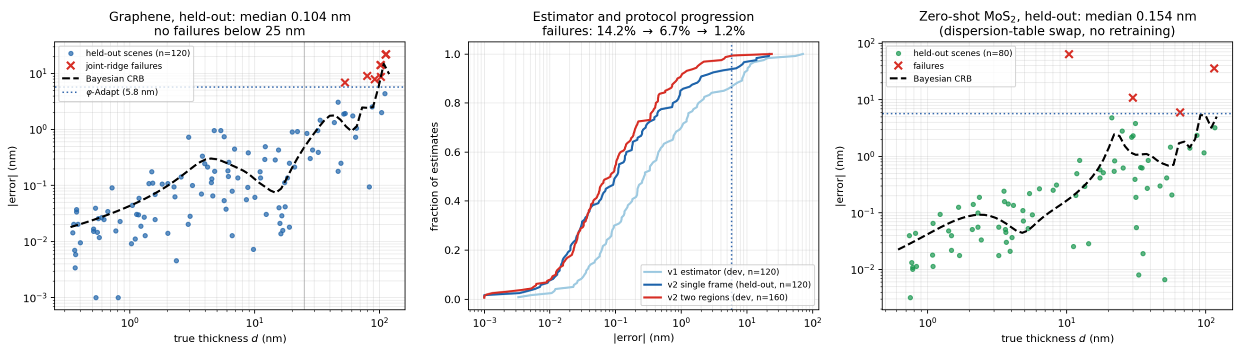

Figure 3: Results. **Left:** graphene errors vs. truth against the Bayesian CRB; all failures (red) lie beyond 25 nm in the joint-ridge domain. **Center:** error CDFs across estimator generations and the multi-region protocol. **Right:** zero-shot MoS_2.

errors 0.030–0.109 nm, worst-case 1.30 nm, and **zero failures in 64 estimates** (16 per illuminant). The explanation is structural, and worth stating precisely. For a fixed stack, each observed channel is a *ratio* of linear functionals of the illuminant: a numerator $\int L(\lambda)R(\lambda)\bar{x}_i(\lambda)\,d\lambda$ and a common normalizer $\int L\bar{y}$. A two-region scene thus depends on L through six numerator functionals and one normalizer; the normalizer is a global scale that the per-channel gains absorb, leaving the illuminant identifiable only up to a projection onto a subspace of dimension at most six. Components of L outside that subspace cannot influence the fit, whatever their magnitude. A basis need only span the *identifiable projection*, not the spectrum itself, which is why a five-component basis plus three gains suffices for a mercury arc whose spectral deviation is 39% unrepresentable: the model does not reconstruct that spectrum, and the unrepresented component simply has little influence on the fitted RGB observables in the regime tested. This also predicts the limit of the argument: acquiring a second frame through a known filter (§5.8) enlarges the identifiable subspace, which is precisely why that protocol works. This is a stronger robustness result than in-family testing could establish. It bears on spectral misspecification specifically, and does not speak to the other ways a real acquisition can depart from the model.

5.3 Layer counting, the field's own metric

Prior systems report discrete layer classes, so for direct comparability we convert our continuous estimates to the nearest integer layer count. Real flakes are integer multiples of the layer height, so for this synthetic layer-count benchmark we *generate* physically discrete thicknesses $d = n \times 0.335$ nm, $n \in [1, 10]$; the ground truth is constructed to match the layer model rather than independently observed (an earlier version of this analysis used the log-uniform continuous thicknesses of the main benchmark, which is not physical and understated accuracy). On 45 such scenes graphene achieves **82.2% exact** layer accuracy and **100% within one layer**, with a median error of 0.15 layers (0.050 nm). Exact accuracy is layer-dependent (100% at 1–2 layers, 90.9% at 3–5, 70.8% at 6–10) because its 0.335 nm interlayer spacing makes adjacent classes increasingly hard to separate as thickness grows. MoS_2, with nearly double the layer height, is correspondingly easier.

The monolayer case, which is what most exfoliation searches are actually looking for, is the strongest result in the paper: median absolute error of **0.0146 nm for monolayer graphene** and **0.0114 nm for monolayer MoS_2**, i.e. 0.04 and 0.02 of a layer respectively. Discrete classification cannot express this: the method does not merely place a monolayer in the right bin, it returns a thickness whose median absolute error is about 4% of a graphene layer and 2% of an MoS_2 layer. That is an error on an inferred quantity, not a claim about the instrument's spatial resolution.

5.4 Comparison to simpler approaches

To establish how much of the performance comes from the full machinery rather than from correct physics alone, we compare against three progressively simpler methods on identical held-out scenes (n=60, seed 201): **B1**, gain-invariant contrast lookup against a curve precomputed at *nominal* nuisances, i.e. the calibrated-contrast approach used in the field but *without* per-setup calibration, and involving no optimization; **B2**, optimizing thickness alone with nuisances pinned at prior means; and **B3**, optimizing thickness and the three channel gains, i.e. conventional white-balance correction without oxide or illuminant estimation. We additionally report **B1-oracle**, the same lookup given the *true* oxide thickness and the *true* illuminant spectrum of each scene. B1-oracle is not achievable in practice, since it presumes exact knowledge of the illuminant SPD. Nor is it a bound on what *any* estimator could attain with known conditions: it is one particular lookup, and another estimator given the same information might do better. What makes it a useful reference is empirical rather than definitional — its median error sits within 5% of the known-nuisance Fisher bound computed independently in §4 (subject to the Gaussian caveat below), so there is little headroom left beneath it, and it is therefore the practical reference against which self-calibration should be judged. B2 and B3 are given 24 multi-start Levenberg–Marquardt fits; their parameter spaces are one- and four-dimensional respectively, so they are not optimizer-limited.

Table 2: Ablation against simpler methods, identical held-out scenes (n=60).

	Median |err|	Mean |err|	Failures	Median, $d < 25$ nm
B1 contrast lookup (field standard)	0.253 nm	5.77 nm	11/60	0.105 nm
B2 fixed nuisances	1.883 nm	11.84 nm	20/60	1.044 nm
B3 gain-only calibration	0.516 nm	5.03 nm	12/60	0.197 nm
Ours (full joint estimation)	**0.099** nm	**1.77** nm	**6/60**	**0.037** nm
B1-oracle (true oxide + illuminant; oracle reference, not achievable; failures over n=45 paired scenes)	*0.020* nm	*0.13* nm	*0/45* *CI [0, 7.9]%*	*0.015* nm

Correct physics alone (B1) is already competitive in median error, which is why calibrated-contrast methods work as well as they do in practice; but its mean error and failure rate are 3–4× and 2× worse, because it has no way to absorb imaging-condition variation. Pinning nuisances (B2) is worst of all, confirming that estimating them matters more than optimizing carefully. Gains alone (B3) recover part of the gap. Full joint estimation improves median error by 2.6× over uncalibrated lookup and 5× over white-balance correction, and halves the failure rate. The ablations therefore indicate that estimating the imaging nuisances, rather than optimizing harder, accounts for most of the gain: B2 and B3 are given 24 multi-start fits over one- and four-dimensional spaces and are not optimizer-limited, so the difference tracks what is estimated rather than how hard it is searched. The four methods differ in more than one respect, so we put this as what the comparison indicates rather than as an isolated cause.

The B1-oracle row frames this result properly and we state it plainly: *if imaging conditions were known, a trivial lookup table would beat our method by a factor of five* (median 0.020 vs. 0.101 nm; on paired scenes B1-oracle is better on 37 of 45). Our method does not compete with knowing the answer; it competes with not knowing it. Reading the three regimes together — uncalibrated lookup 0.253 nm, ours 0.101 nm, oracle-calibrated lookup 0.020 nm — self-calibration recovers roughly 65% of the gap between the uncalibrated lookup and the oracle-calibrated lookup, while requiring no reference samples, no known-thickness standards and no per-setup procedure under the stated optical model. We measure the gap against the oracle lookup rather than against ‘perfect calibration’ in the abstract, since the former is a benchmark we ran and the latter is not a quantity we have. This comparison also provides an *implementation-independent consistency check on the identifiability theory of §4*, which matters because the bounds and the estimator share no code path (the two rest on the same forward physics and the same scenes, so the check is of the implementation, not of the theory’s premises): B1-oracle is a 600-point table lookup, while the bounds come from automatic differentiation of the forward model. For zero-mean Gaussian errors median$|e|$ = $0.6745\,\sigma$. B1-oracle’s median error of 0.0204 nm therefore corresponds, *under that assumption*, to $\sigma = 0.0302$ nm against a predicted oracle bound of 0.0289 nm, an agreement of 1.05×. We have not established that the oracle error distribution is Gaussian, and a CRB bounds a variance while the reported statistic is a median absolute error, so this is corroboration of the Fisher computation rather than a demonstration of efficiency. Our method’s median of 0.1014 nm implies $\sigma = 0.150$ nm against a full-nuisance bound of 0.233 nm, a ratio of 0.64, consistent with the per-scene analysis in §5 and with the shrinkage expected of a MAP estimator under informative priors. The residual 35% is the remaining gap to the oracle lookup. Inferring the imaging conditions is the obvious contributor, and a spectrally richer acquisition (§5.8) is designed to reduce it, but we have not separated that contribution from differences of estimator form, lookup discretization and prior shrinkage, so we do not attribute the whole of it to nuisance inference.

5.5 Coverage: alternate substrate and a transparent material

The main benchmark uses 285 nm oxide and two absorbing materials. We add the other standard wafer and the field’s acknowledged hard case. **90 nm oxide** (graphene, n=40): median 0.052 nm, mean 1.30 nm, 3/40 failures (7.5%, 95% CI [1.6, 20.4]), 0/28 below 25 nm. **hBN on 285 nm oxide** (n=40), which is transparent in the visible (k=0) so that contrast arises from phase alone and which MaskTerial identifies as its difficult low-contrast case: median 0.113 nm, mean 0.531 nm, **0/40 failures** (95% CI [0, 8.8]%). A physics-based estimator is not disadvantaged by low contrast in the way a contrast classifier is, because the thickness information resides in the interference phase rather than in contrast magnitude.

5.6 Sensitivity to the assumed optical constants

The forward model treats the material’s dispersion as known, yet tabulated n, k for 2D materials disagree between sources by more than a few percent, and graphene’s constant-index model is itself an approximation. We render scenes with n and k scaled by $(1+\varepsilon)$ while the estimator retains the nominal values (n=30 per setting). At ε=2%: median error 0.407 nm, 0/30 failures, sub-25 nm median 0.079 nm. At 5%: 1.076 nm, 5/30 failures, sub-25 nm 0.183 nm. At 10%: 1.785 nm, 10/30 failures, sub-25 nm 0.349 nm.

Two features matter. The error is *systematically signed* (median +0.095, +0.180, +0.328 nm), as expected: an overestimated refractive index is compensated by an overestimated thickness, so dispersion error appears as a calibration offset rather than as noise. And the thin regime is far more robust than the thick one, staying sub-nanometre even at 10% dispersion error, while failures concentrate above 25 nm where ridges already exist. Practically this means the method inherits the accuracy of its dispersion table in the thick regime and should be paired with material-specific tabulated data rather than constant-index approximations when absolute accuracy beyond a few nanometres is required. Estimating n, k jointly is possible in principle given enough regions sharing one material, and is the natural extension.

5.7 Sensitivity to imperfect segmentation

Since we assume given regions (§1), we quantify the cost of mask error by mixing a fraction α of substrate signal into the flake region, as a dilated or leaky mask would (graphene, n=40 per setting, seed 211). Median error rises from 0.130 nm at α=0 to 0.135 (α=2%), 0.226 (5%), 0.280 (10%) and 0.502 nm (20%), with failures 2, 2, 2, 1 and 4 of 40 (95% CI at α=20%: [2.8, 23.7]%). Degradation is graceful and sub-nanometre accuracy survives 20% contamination, but an end-to-end system would inherit its segmenter’s error on top of this, and we have not measured that composition.

5.8 Acquisition protocols for the thick regime

All remaining failures pair the truth with a joint-ridge partner beyond 25 nm. Because the ambiguity is information-theoretic for a single frame, the remedies are acquisition protocols, each derived from the theory and each costing almost nothing in practice. We report these on 25–80 scenes; they establish large effects, not the absence of failure, and the confidence intervals below should be read as the operative claim:

Multi-region (mode lattice). On 80 two-region scenes (160 estimates; development seeds, not held-out): median 0.084 nm, failures 9.4% → **1.3**% overall and 41.7% → **5.6**% in the hard d>25 nm regime relative to the v1 joint search, at 2.7× lower cost. Three regions, in an all-hard-regime stress test (every $d \in [28, 120]$ nm): **0/60 failures** (95% CI [0%, 6.0%]), median 0.27 nm, versus 7/60 for single-frame estimation of the same regions.

Known-filter two-frame protocol. A second photograph of the same field through a known color filter (an inexpensive gel; we model a smooth long-pass with 0.12 floor), with illuminant coefficients *shared* across frames and per-frame gains, converts two RGB frames into coarse spectroscopy: hard-regime failures 3/25 → **0/25** (95% CI [0%, 13.7%]; the sample is small and we report it as strong evidence of a large effect, not as a demonstration of exactly zero failure probability). The control experiment matters: two frames under two *unknown* illuminants barely help (5/25 → 4/25), because each frame's unknown spectrum hands the wrong ridge fresh nuisance freedom — the spectral change must be *known*.

Profile-triggered acquisition. The second frame is requested only when the profile posterior shows any rival basin within a wide trigger window (40 units). On hard-regime scenes this fired 11/25 times and caught the confidently-wrong cases that mode-count triggers miss entirely (§4.1); on the standard benchmark the vast majority of scenes are certified unimodal and need one frame.

5.9 Numerical aperture: the dominant systematic, and a hard requirement

Real objectives illuminate over a cone. We implement oblique-incidence TMM (s/p characteristic matrices, pupil-uniform quadrature in $\sin^2\theta$), validated to machine precision against [7] at angle. This turns out to be the single most consequential modeling choice in the paper, and we report it plainly because our own first analysis of it was wrong.

NA sets the contrast-optimal substrate. The oxide thickness maximizing monolayer contrast at 550 nm moves from 88/276 nm at normal incidence to 90/286 (NA 0.55), 94/296 (NA 0.75) and 96/304 nm (NA 0.90). Flake-hunting objectives are typically 50–100× with NA 0.55–0.9, and at those apertures the model reproduces the community's standard 90 nm and 285–300 nm wafers. Magnitudes agree too: on 300 nm oxide at 550 nm our NA 0.9 contrast is 0.090, inside the 0.09–0.12 range reported by Blake et al. [1] using a 100×/0.9 objective, whereas the normal-incidence value of 0.063 is well below it. The standard wafers are not folklore; they are correct for the optics actually used, and reproducing them at the right NA is an independent validation of our forward model.

Mismatched NA is catastrophic, including where we otherwise never fail. Rendering truth at NA 0.90 and estimating with a normal-incidence model gives median error 1.53 nm, mean 5.56 nm, 11/30 failures, and **5/21 failures below 25 nm** — the only setting anywhere in this paper that produces sub-25 nm failures. At NA 0.55 the same mismatch is benign (median 0.097 nm, no failures), which is why a first pass at moderate aperture badly understated the risk. Matching the forward model to the objective removes the problem entirely: at NA 0.90 with an NA-aware estimator, median error is 0.026 nm with 0/12 sub-25 nm failures.

Requirement, not a refinement. The numerical aperture of the objective must be known and built into the forward model. This is a one-line change and the value is engraved on every objective, but omitting it silently destroys thin-regime accuracy. Pupil quadrature needs ≥ 6 nodes for metrology-grade use, at roughly 6–10× the forward-model cost.

5.10 Robustness

8-bit imagery. Real micrographs arrive quantized. Encoding each pixel to sRGB, rounding to 8 bits, adding read noise and averaging over the region reproduces the continuous-input result: median 0.051 nm, sub-25 nm median 0.042 nm, 0/30 failures (n=30, 100 flake pixels). Quantization is dithered away by read noise and is not a limiting factor. We note one caution: averaging 1000 rather than 100 pixels did not improve the median (0.231 vs. 0.051 nm on n=30), consistent with a small quantization-induced bias floor that further averaging cannot reduce; the sample is small and this deserves confirmation on real data.

With the noise model mis-set by 2× and 4× (truth noisier than assumed): medians 0.087/0.120 nm, zero failures. With the wafer mislabeled by 3σ (true oxide $\mathcal{N}(300, 5)$ nm against a 285 nm prior): median 0.128 nm, 1/20 failures — self-calibration recovers the true oxide from the data and overrides the label.

5.11 Negative results and ablations

We report what failed, verified pairwise on identical scenes. (1) *First-order multi-start* (32 Adam [24] restarts; "v1" in Table 1): the accuracy gap to the continuation estimator traces to underconvergence and non-exhaustive mode discovery. (2) *Closed-form gain marginalization* (Rao–Blackwellization): mathematically exact, empirically worse (mean error 2.06 vs. 1.01 nm, two new failures) and slower — profiling gains degrades the Gauss–Newton geometry and permits negative gains. (3) *Richer illuminant basis alone* (5 vs. 3 components, before the positivity constraint and dual-start solves): a wash — fixed 6 failures, introduced 7; added flexibility helps wrong ridges as much as right ones. The accuracy gains in this paper come from the diagnosed physical constraints, not from added capacity. *Statistical caveat:* we tested these design decisions by paired comparison on 120 scenes, where failures are rare events. Only the aggregate first-order-to-continuation change is significant (11 failures fixed, 0 introduced; exact McNemar p=0.001); the individual accept/reject decisions above rest on differences of 1–7 failures that are *not* statistically distinguishable from noise ($p \geq 0.13$). We report them as engineering judgements supported by mechanism, not as established effects. (4) *Adaptive filter selection*: choosing the frame-2 filter by maximizing predicted mode separation was unreliable — on the one scene where choice mattered, the score's top-ranked filter failed (101.1 nm vs. truth 94.0) while its worst-ranked filter succeeded (94.7 nm), because the score is computed from drifted-nuisance predictions (§4.1). The validated protocol is the trigger plus a fixed known filter; ridge-aware selection is future work.

5.12 Real-data evaluation on public micrographs

We evaluate on the public MaskTerial `GrapheneL` dataset [3]: real reflected-light micrographs of exfoliated graphene with per-pixel layer annotations, acquired on a 20× objective by a different laboratory with no involvement from us. Flake and substrate regions are taken from the released semantic masks; no other preprocessing is applied.

The bare-substrate term is uninformative on these images, and diagnosing that is the first result. The wafer here is strongly coloured, as the physics requires: the bare substrate reads a median 8-bit RGB of $[113, 107, 151]$ over the 1362 test images, blue-dominant, against a forward-model prediction of $[0.110, 0.098, 0.177]$ in linear RGB at 90 nm oxide, likewise blue-dominant. The comparison is of hue and not of magnitude, since the absolute scale is exposure and is arbitrary. The difficulty is not the substrate's colour but the three free channel gains of (3): with three unknowns and the three observed substrate channels, they reproduce *any* substrate colour exactly, so the bare-substrate term carries no information about (t_{ox}, c) once the gain prior is unable to hold them near unity. On these images the fitted exposure alone sits near 4, roughly 14σ from a log-gain prior of width $\sigma_{\log g}$=0.1 centred at unity, and the joint estimate is correspondingly ill-conditioned. The self-calibration consistency check of §5.4 flagged this before any ground truth was consulted, which is the intended behaviour: a metrology method should refuse data it cannot interpret.

Restricted mode restores identifiability. Pinning the illuminant and treating the oxide as a single global constant per dataset leaves the substrate determining the three gains exactly (3 equations, 3 unknowns) and the flake determining one thickness from three observations. The oxide is calibrated once on the *train* split and the *test* split is then evaluated without further adjustment. This is the physical analogue of the adaptation baselines perform: MaskTerial requires 5–10 labeled images per material [3]; we require one scalar.

Two acquisition parameters are recovered from the images themselves. Sweeping oxide thickness over 60–320 nm at 5 nm steps produces a single sharp global minimum in color-fit residual at 90 nm (neighboring candidates are 2.5–9× worse), recovering a value consistent with the nominal ∼90 nm graphene wafer specification without being told it; refining the sweep around that minimum gives the 92 nm used in

Table 4. Separately, testing sRGB versus linear decoding showed the released images are radiometrically linear rather than sRGB-encoded, a fact not stated in the dataset documentation: assuming sRGB inflated recovered thickness by a factor 2.19 and produced a residual that grew with thickness, while linear decoding simultaneously reduced the residual, moved the recovered oxide to nominal, and collapsed the scale error to 0.6%. Four independent diagnostics moving together is what distinguishes a physical correction from a tuned parameter.

Result on the held-out test split (n=200 flakes, 1–4 layers) is given in Table 3:

Table 3: Real-data evaluation, MaskTerial `GrapheneL` test split. Oxide calibrated on the train split only; no labeled examples used for training. Intervals are bootstrap 95% CIs over 4000 replicates, as in Table 4; the Clopper–Pearson intervals of Table 1 apply to failure counts only.

Exact layer agreement	**97.0%** (95% CI [94.5, 99.0])
Within one layer	**100.0%**
Recovered layer height	0.3380 nm/layer, bootstrap 95% CI [0.3316, 0.3449] (literature 0.335; error +0.9%)
Intercept	+0.029 nm
Correlation with annotation	$r = 0.9847$
Flakes with poor individual color fit	4/200
Per-class exact agreement	98.4 / 95.7 / 95.1 / 100.0% (1/2/3/4 layers)

The recovered slope is a physical thickness scale, not an ordering of the supplied labels. This is the substantive claim of the real-data experiment and we state it explicitly, because the natural objection is that agreement with optically-derived annotations could reflect nothing more than reproducing their rank order. Across held-out real images the inferred thickness rises by 0.3380 ± 0.0043 nm per annotated layer on `GrapheneL` and 0.3388 ± 0.0054 nm on `GrapheneM`, agreeing with graphene's independently known 0.335 nm interlayer spacing to 0.9% and 1.1% respectively. A method that had learned only to rank the annotations would have no reason to reproduce that constant at all. (These two releases overlap and are not independent acquisitions; see below.) The interlayer spacing enters the inference at no point: thickness is estimated as a continuous quantity in nanometres, the annotations are integer class indices, and the slope relating them is fitted afterwards. The 0.335 nm figure appears in this paper only as the literature value the fitted slope is compared against, and as the divisor used to report layer counts. What this does *not* establish is per-flake accuracy against an independent physical measurement, which requires AFM or Raman and remains the necessary next test.

What this does and does not establish. The annotations are layer classes assigned from optical images, so agreement demonstrates that we reproduce the community's labeling on the community's data, not agreement with an independent physical measurement; AFM or Raman validation remains future work. The residual still grows with thickness ($0.036 \rightarrow 0.099$ across classes) and the log-log exponent is 0.977 ± 0.011, marginally excluding pure proportionality, so a small unmodelled nonlinearity remains in their pipeline. Results are reported for one material on one dataset over 1–4 layers.

Transfer across the designed substrate-variance series. MaskTerial ships graphene as `GrapheneL/M/H`, which are not three acquisitions but three samplings of one pool of named exfoliation runs, constructed to span approximately 5, 10 and 20 nm of oxide about a nominal ∼90 nm wafer [3], and overlapping by 162 byte-identical images between `L` and `M`. We apply an identical command to each: oxide and transfer function re-derived independently from each dataset's own images, no labeled examples, no per-dataset tuning. Table 4 reports the result.

The same scale is recovered on the larger draw. The objection above applies equally here to any evaluation scored against annotations, and the data answers it directly: across the held-out real images the inferred thickness increases by 0.3370 ± 0.0032 nm per annotated layer (n=600), in agreement with graphene's independently known 0.335 nm interlayer spacing to 0.6%. A method that had learned only to reproduce the community's labeling would order flakes correctly but carry no obligation to recover that constant; the constant is a property of the crystal, is nowhere supplied to the estimator, and enters only through the tabulated optical dispersion. We are careful about what this does and does not establish: it shows the *scale* of the recovered thicknesses to be consistent with the independently known interlayer spacing — a value inside the interval is not a demonstration that the true value equals it — and not that any individual flake

Table 4: Transfer across the MaskTerial graphene substrate-variance series. L, M and H denote low, medium and high substrate-thickness variance: the releases were built to span ∼5, ∼10 and ∼20 nm of oxide about a nominal ∼90 nm wafer [3], and draw on a shared pool of exfoliation runs rather than being independent acquisitions, so the fitted oxide is a mixture-weighted compromise rather than a wafer measurement. Oxide recovered from each dataset's own images without labels, by minimizing color-fit residual over a 60–320 nm sweep; transfer function selected automatically from the images; nothing tuned per dataset. The n=600 row reuses the oxide recovered from the n=200 GrapheneL scan and is a larger draw from that same acquisition; the three n=200 rows are independent recoveries. Intervals are bootstrap 95% CIs (4000 replicates).

Dataset	Oxide	Exact agreement	Recovered layer height (nm)	r
GrapheneL (n=600)	92 nm	95.3% [93.7, 96.8]	0.3370 [0.3310, 0.3439]	0.974
GrapheneL (n=200)	92 nm	97.0% [94.5, 99.0]	0.3380 [0.3316, 0.3449]	0.985
GrapheneM (n=200)	96 nm	93.5% [90.0, 96.5]	0.3388 [0.3284, 0.3489]	0.976
GrapheneH (n=200)	98 nm	76.5% [70.5, 82.0]	0.2904 [0.2694, 0.3102]	0.892

has been measured against an independent instrument. Per-flake validation against AFM or Raman remains the necessary next experiment (§6).

The recovered spacing is consistent across the series to within sampling noise. Graphene's 0.335 nm lies inside the confidence interval for GrapheneL and GrapheneM, and every pairwise bootstrap comparison among them is consistent (L_{600} vs. M: −0.0018 nm, CI [−0.0134, +0.0110]). These are not independent acquisitions and we do not claim them as such: GrapheneL and GrapheneM share 162 images, byte-identical where they overlap, and each release is a sampling from a shared pool of exfoliation runs. The fitted values of 92, 96 and 98 nm are accordingly mixture-weighted compromises over the nine or ten runs each release contains rather than three wafer measurements, and the upward drift with the series' designed oxide spread is the expected behaviour of a single-oxide fit to a broadening distribution. What the agreement does establish is that the recovered scale is physical: 0.335 nm is supplied to the estimator at no point.

Two robustness checks accompany this. The bootstrap CI half-width on layer height falls from ±10.9% at n=25 to ±2.0% at n=600, scaling as $n^{-1/2}$, which is what sampling uncertainty alone would produce and means more flakes would tighten the interval further. It does not exclude a systematic bias: a fixed offset and an $n^{-1/2}$ sampling term coexist without disturbing that scaling. Varying the mask-erosion depth from 1 to 6 pixels changes the recovered layer height by under 2% on both GrapheneL (0.3355–0.3402) and GrapheneM (0.3341–0.3410), within the sampling interval, so the one free preprocessing parameter is not driving the result.

GrapheneH differs from all three with CIs excluding zero, reading 13% low with the deficit concentrated in its thicker classes; within-one-layer agreement remains 97.5%, so ordering is preserved and only absolute scale degrades. We report it rather than the best two. Notably its color-fit residual is the *lowest* of the four, so the self-diagnosis does not flag it — bounding what that check can be trusted to catch: it detects images the physics cannot explain, not annotations that disagree with the physics.

The single global oxide is what fails on GrapheneH, and fitting per exfoliation run shows it. The releases record the exfoliation run behind every image, so the assumption can be removed rather than argued about: we repartitioned the corpus on that variable, deduplicating images shared between releases, and ran the *unrestricted* estimator of §3.2 on each run — illuminant free, nuisances fitted per flake, no labels used at any point — taking each run's oxide as the median over its own flakes. This is the joint mode the restricted analysis above sets aside, applied where its single-wafer premise actually holds. Twenty-one runs, 739 flakes, Table 5. Sixteen of them return substrate thicknesses from 89.2 to 101.6 nm with a median of 93.47 nm, a span of 12.4 nm. We are careful about the resolution this buys: a single run's oxide is determined to a median absolute deviation of 3.6 nm (range 0.9–4.8), so the ends of that span are separated by roughly two standard deviations while neighbouring runs are not individually resolved. What the measurement establishes is that a spread of order 10 nm is present and recoverable from the images with no annotations, the magnitude the dataset authors report designing into the series [3], not that each run's wafer has been measured. Four further runs carry the white-balance signature described above for hBN_Thin — fitted exposure 6.2–7.6 against 1.7–5.0 for the rest — and sit 15 nm above the others at 108.5 nm, which we report but cannot resolve between a different wafer and a displacement caused by the vacuous substrate term. One run is refused outright by the self-calibration consistency check of §5.4, its fitted oxide scattering by 33.7 nm across a single wafer. `GrapheneH`'s 13% deficit is therefore consistent with representing that spread by a single number, rather than being unexplained; we have not shown that the spread accounts for its exact magnitude. The remedy is available from metadata the release already ships.

Table 5: Substrate oxide recovered per exfoliation run, from the images alone with no annotations, after repartitioning the graphene corpus on the run identifier in its own metadata and deduplicating images shared between releases. Scatter within a run is median absolute deviation over that run's flakes; the range is across runs. The white-balanced group is identified by fitted exposure alone, without reference to annotations.

Group	Runs	Flakes	Median oxide	Range	Within-run scatter
Consistent	16	545	93.47 nm	89.2–101.6 nm	0.9–4.8 nm
White-balanced	4	160	108.46 nm	107.5–109.3 nm	5.5–7.5 nm
Refused by §5.4	1	34	—	—	33.7 nm

Materials where the method refuses, and one where the labels are the puzzle. Applying the same command to the released TMD datasets, the color-fit criterion rejects both: `WSe2` (residual 0.336, recovered layer height −21.8%, 120/135 flakes individually flagged) and `MoSe2` (residual 0.125, +64.7%, 35/95 flagged). These are correct refusals. We stress that `MoSe2` simultaneously reports 100% "exact agreement", because with only two label classes and a systematic scale error, rounding lands correctly — a demonstration that the field's standard classification metric can appear perfect while the underlying physics is wrong, and an argument for reporting a physical consistency check alongside it.

`hBN_Thin` is a different and more interesting case, and two things distinguish it. Its wafer reads neutral grey (median RGB $[156, 159, 154]$, inter-image IQR $[2.0, 2.2, 0.2]$) where the physics predicts a strongly coloured substrate, indicating a camera white-balanced on the substrate; there the bare-substrate term is vacuous by construction rather than merely ill-conditioned. And it produces the *cleanest* residual of any dataset we ran (0.0403, a sharp isolated minimum at 80 nm) with $r = 0.94$, yet a recovered layer height of 1.68 nm against hBN's 0.333 nm spacing. The estimates for label classes 1, 2, 3 are 1.25, 3.0 and 4.7 nm, a near-constant ratio of $\sim$5 with high linearity. A physics failure does not produce a clean residual and a linear response; the more parsimonious reading is that these class indices are thickness *tiers* rather than monolayer counts, which would be consistent with device-grade hBN thicknesses. The released metadata contains only UUID mappings and does not define the classes, so we advance this as a hypothesis rather than a finding, and exclude hBN from our quantitative claims.

We are aware that "the labels must mean something else" is a convenient explanation whenever a method disagrees with ground truth, and that an explanation always available to us is worth little. We therefore state what would settle it. If the classes are monolayer counts, our estimates are wrong by a factor of five and no acquisition change should repair them. If they are thickness tiers, three predictions follow: an AFM step height on any labeled flake should read near our estimate (1.25, 3.0 or 4.7 nm) rather than near $n \times 0.333$ nm; thickness within a single class should be dispersed rather than clustered, since a tier spans a range; and re-running with the class-to-thickness map implied by our estimates should reduce the residual, whereas re-running with monolayer counts should not. The first of these is decisive and requires one AFM scan. Until it is done we report hBN as unresolved rather than as either a success or a failure, and it enters none of our accuracy claims.

6 Limitations

Real-data validation (§5.12) uses optically-derived annotations rather than an independent physical measurement, covers one material over 1–4 layers, and required empirically determining that the released images are radiometrically linear, a fact absent from the dataset documentation. The three graphene releases are a designed substrate-variance series drawn from a shared pool of exfoliation runs rather than independent acquisitions — `GrapheneL` and `GrapheneM` share 162 byte-identical images — so they probe robustness to oxide spread rather than transfer between laboratories, and genuine cross-laboratory transfer remains untested. The restricted mode's single global oxide is violated by construction on the high-variance release; fitting per exfoliation run removes that assumption and recovers the corpus wafer distribution (Table 5), but does so on the same images and annotations and so inherits the same limitation. The comparison to published numbers is range- and metric-matched but not protocol-matched (different images, and our pipeline is given flake regions rather than segmenting them). Baselines were not re-run under our protocol. Protocol claims in §5.8 rest on samples of 25–60 scenes; confidence intervals are reported and the upper bounds are not negligible. Throughput is a practical disadvantage: at roughly 1.5 s per flake on CPU, a wafer scan containing several hundred flakes takes minutes, against near-real-time for feed-forward classifiers; the profile sweep is trivially parallel across flakes and the cost is dominated by the 140-point nuisance solve, so this is an engineering rather than fundamental limit. The objective's numerical aperture must be known and modeled; a normal-incidence approximation is adequate only up to NA ≈ 0.55 and fails badly at NA 0.9 (§5.9). Absolute accuracy in the thick regime inherits the accuracy of the assumed dispersion data (§5.6); dispersion error enters as a signed calibration offset rather than as noise. Efficiency statements rest on linearized prior-augmented bounds compared per scene, which is a diagnostic rather than a rigorous efficiency proof. Design-decision comparisons are underpowered for rare failure events, as stated in §5.11. The multi-region and protocol experiments use development seeds and samples of 25–80 scenes; only the single-frame benchmark is fully held-out. Beyond $\sim$25 nm, single-frame RGB observation is ridge-limited for any method that must infer imaging conditions from the image itself; a method handed the true oxide and illuminant is not so limited (Table 2), which locates the difficulty in the calibration rather than in the optics; our protocols substantially reduce it at the cost of one extra frame or one extra region, on the sample sizes noted above. Contrast saturation ultimately bounds all optical methods in the very thick regime, where AFM and Raman remain the arbiters.

7 Conclusion

Placing differentiable thin-film optics inside the inference loop turns 2D-flake thickness estimation from bin classification with learned domain patches into continuous, self-calibrating physical parameter estimation. Measured against prior-augmented per-scene bounds, the resulting estimator is not the limiting factor on accuracy where its forward model is correct — a diagnostic rather than a proof of efficiency, since a MAP estimator under informative priors is deliberately biased — transfers across materials with a table swap, knows — and can repair, with one cheap extra measurement — its own ambiguous domain, improves on the field-standard calibrated-contrast approach by 2.6× in median error under matched conditions and on the published state of the art by 2× in mean error under a range- and metric-matched comparison (given segmented regions in both of our settings), with no labeled training images (the illuminant basis is built from eleven published illuminant spectra, which we regard as calibration data rather than training data). Its residual error is not attributable to the inference: measured against per-scene bounds the estimator is at or below them, and the remaining limits are properties of the measurement itself — the ridge structure of the thick regime, and the requirement that the objective's aperture and the material's dispersion be known.

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Reproducibility. All code (roughly 600 lines of core modules, 950 including every experiment script), optical-constant provenance, and experiment scripts with seeds are released. They are available at https://github.com/bpetrie36/flakedepth (accessed August 2026).

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